Averages and Independent Probability

Concept

Averages describe a typical value. Probability describes how likely an event is.

Centre

Mean = sum/count; median = sorted middle; mode = most frequent.

Outlier

An extreme value affects the mean more than the median.

Probability

Favourable/total outcomes, when outcomes are equally likely.

Independent

P(A and B) = P(A)P(B); P(not A) = 1 - P(A).

Example 1

Find mean, median and mode of 2, 3, 3, 4, 18.

Solution

1. Mean = 30/5 = 6. Median = 3. Mode = 3.

2. The value 18 pulls the mean up; the median better represents a typical value here.

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Example 2

Toss two fair coins. Find the probability of exactly one head.

Solution

1. Sample space: HH, HT, TH, TT.

2. Successes: HT and TH → $P = 2/4 = 1/2$.

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Visual notes | two-stage sample space

	H	T
H	HH	HT
T	TH	TT

Rows show the first fair coin; columns show the second. Four equally likely ordered outcomes remain after the labels are removed. Exactly one head occurs in HT or TH, so its probability is 2/4 = 1/2.

A tree gives the same result: each first toss branches into H or T, and each of those branches splits into H or T again. Multiply probabilities along a path; add probabilities of disjoint successful paths.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find mean, median and mode for 4, 5, 5, 6, 10.

2. Toss two fair coins. Find P(at least one head).

3. Roll two fair dice. Find P(both show 6).

4. Why should a survey report more than just its mean?

5. Are two card draws without replacement independent?

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Practice answers | check after trying

1. Mean = 30/5 = 6; median = 5; mode = 5.

2. 1 - P(TT) = 1 - 1/4 = 3/4.

3. (1/6)(1/6) = 1/36.

4. Different distributions and outliers can have the same mean; sample size, spread and context help interpret it.

5. Generally no: removing the first card changes the possible outcomes and probabilities for the second draw.